How Much PENCIL Can You Afford? Compute-Allocation Laws for Link Prediction

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I. TWO-STAGE RETRIEVE-THEN-RERANK PIPELINE

Candidate Pool — N pairs (Pool) or 501 / query (HeaRT) → **Stage 1 · Adamic–Adar** — CPU <1 ms · 0% GPU ranks & shortlists top k → **Stage 2 · PENCIL** — Transformer re-scores top k with budget s

TWO INFERENCE KNOBS (NO RETRAINING): **k** = shortlist re-ranked (0–100%) · **s** = subgraph hops×fanout

II. FREE-GPU REPRODUCTION & AUDIT (PAPER USED 4–8× A100)

44.00
Cora Test MRR
Paper: 42.23 · 1×T4, 3.2h

50.46
Citeseer Test MRR
Paper: 47.51 · 2×T4, 1.0h

112 Runs
4 Regimes Audited
HeaRT AA exact · $\Delta < 10^{-7}$

III. FOUR EMPIRICAL COMPUTE-ALLOCATION FINDINGS

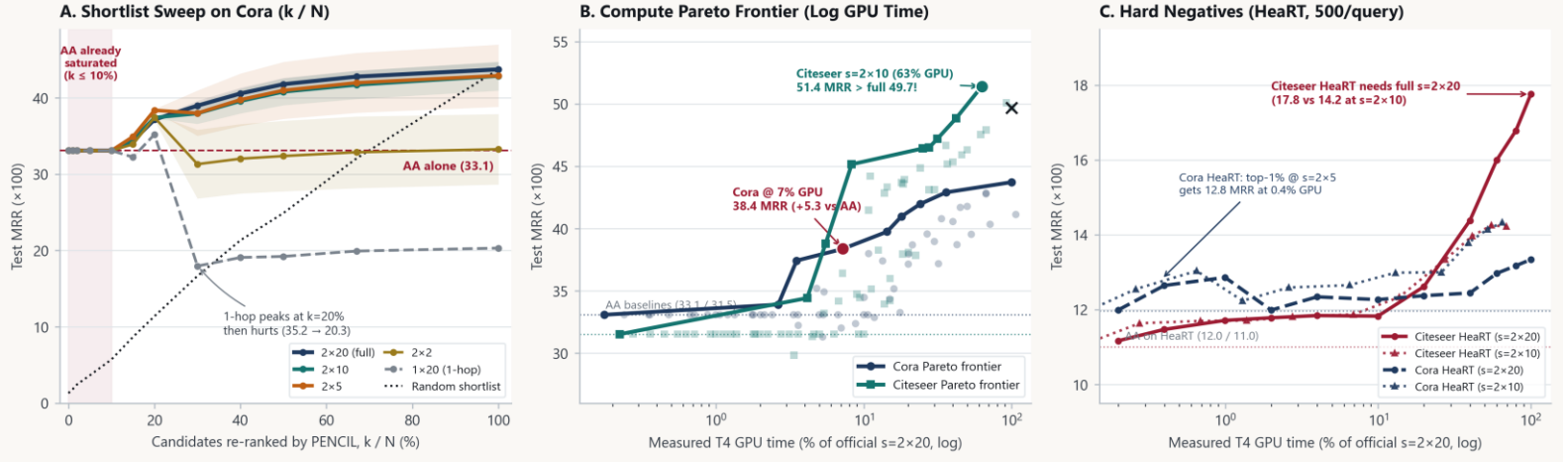
01 — Official s = 2×20 over-reads in 3 of 4 regimes
Halving fanout to s = 2×10 (~65% GPU, 40% less VRAM) matches Cora (42.9 vs 43.7) and beats full PENCIL on Citeseer Pool (51.4 vs 49.7) and Cora HeaRT (14.3 vs 13.3). Only Citeseer HeaRT needs full 2×20 (17.8 vs 14.2).

02 — Frugal hybrid (≤10% GPU) beats heuristics in all 4 tests
Choosing (k, s) on validation alone under a 10% compute cap lifts Test MRR over Adamic–Adar everywhere: 33.1 → 38.4 on Cora (7% GPU) and 11.0 → 13.8 on Citeseer HeaRT (8% GPU).

03 — Top-10% re-ranking is wasted; PENCIL pays off in the 0-CN tail
AA's top decile is already true links, so re-ranking k ≤ 10% changes nothing. PENCIL earns its FLOPs where AA is blind: 57.3% of true Cora test links share zero common neighbours.

04 — Cross-graph budget rules beat noisy local validation
Applying Cora's validation-chosen (k, s) unchanged to Citeseer Pool beats Citeseer's own validation pick at every cap ≤ 50% (44.9 vs 37.9 MRR at 25% cap) because small-graph validation is noisy.

IV. SHORTLIST SWEEP, PARETO FRONTIER & HARD-NEGATIVE (HeaRT) REGIMES



V. VALIDATION-SELECTED OPERATING POINTS & CROSS-GRAPH TRANSFER (TEST MRR ×100, MEAN ± STD)

Operating Point / Policy	Cora Pool (527 shared neg)	Citeseer Pool (455 shared neg)	Cora HeaRT (500 hard/query)	Citeseer HeaRT (500 hard/query)
Chance level (random ranking)	1.3	1.5	1.4	1.4
Adamic–Adar (AA) alone [0% GPU, <1 ms CPU]	33.1 (31.9 avg)	31.5 (29.4 avg)	12.0 (11.9 avg)	11.0 (10.8 avg)
★ **Frugal Hybrid [val-picked ≤ 10% GPU cap]**	**38.4 ± 0.1 (7%)**	**34.7 ± 2.6 (9%)**	**12.8 ± 0.2 (0.4%)**	**13.8 ± 0.5 (8%)**
★ **Cross-Graph Rule [Cora → Citeseer, 25% cap]**	**42.0 ± 3.9 (24%)**	**44.9 (beats own 37.9)**	**12.9 ± 0.1 (23%)**	**13.8 (21% GPU)**
★ **Half-Fanout PENCIL [s = 2×10, all candidates]**	**42.9 ± 1.9 (67%)**	**51.4 ± 0.7 (63%)**	**14.3 ± 0.0 (65%)**	**14.2 ± 0.0 (69%)**
Official Full PENCIL [s = 2×20, 100% GPU, 2.2 GB]	43.7 ± 0.9 (100%)	49.7 ± 0.2 (100%)	13.3 ± 0.1 (100%)	**17.8 ± 0.1 (100%)**

Protocol note: Parentheses in AA row show ties-averaged MRR (matching published HeaRT tables: 31.87, 29.37, 11.91, 10.82); degree tie-break lifts AA to 33.1 / 31.5. Peak VRAM under HeaRT drops from **2.219 MB (s=2×20) → 1.347 MB (s=2×10) → 558 MB (s=2×2)**.

ROADMAP → NOV 26
Final Report · Video · Code

1. **Training-axis compute law:** 93% of final Cora Test MRR is reached by epoch 42 of 150 (28% of training budget), after which validation/test curves plateau.
2. **Remaining deliverables:** Fit closed-form (k, s) selector on validation · latency + VRAM Pareto frontier · reproducible suite & 112 score dumps at `github.com/Ask-812/da4022-pencil-budget`